

Spotify Genre Classification Report

Overview and data preparation

For this project, I built a classification model to predict a song's genre from Spotify audio features. I started by checking the structure of the dataset before doing any modeling because I wanted to make sure the data was being handled correctly. The original file had 50,005 rows, but 5 of those rows were blank or missing the genre label, so I removed them. After cleaning, the dataset had 50,000 usable songs with 10 balanced genres. I also converted tempo into a numeric column because some values were stored as a question mark instead of a number. Since tempo had 4,980 missing values after this conversion, I didn't delete those rows. Instead, I filled the missing tempo values with the median inside the pipeline, making sure this step was learned only from the training data.

I followed the required train/test split by randomly selecting 500 songs from each genre for the test set and using the remaining 4,500 songs from each genre for training. This gave me 45,000 training songs and 5,000 test songs. I also checked for overlapping indexes between the training and test sets, and there were 0 overlapping rows. This was important because even small leakage would make the final AUC look better than it actually is.

Model design choices

For the first model, I used Logistic Regression as a baseline. The baseline used the numeric audio features and popularity, plus the categorical variables key and mode. I did not use instance_id because it is only an identifier. I also left out artist_name and track_name because they would require text processing, and I wanted the main model to focus on the Spotify audio features rather than memorizing names or artists.

After building the baseline, I added PCA and KMeans features to see if they could help the model pick up more patterns in the data. For PCA, I only used the numeric features because key and mode are categorical, and I did not want to treat them like normal continuous variables. I chose 6 principal components using an 80 percent variance threshold. These 6 components explained about 80.38 percent of the variance, which felt like a good balance between reducing the number of dimensions and still keeping most of the structure in the data.

For clustering, I tested KMeans with k values from 2 through 15 and looked at both the elbow plot and silhouette scores. The silhouette score was highest at k = 2, but I did not think that was the best choice because it would split the songs into only two very broad groups. Since the dataset has 10 genres, I

wanted a cluster number that captured more detail without making the model too complicated. I chose $k = 6$ because the elbow plot showed that inertia was still decreasing, but the improvement started to slow down around that point. I, then, added the KMeans cluster label as another categorical feature for the final classifier.

Model results and interpretation

The final model I chose was a Random Forest classifier using the original features, PCA components, and KMeans cluster labels. I also tested a Linear SVM with the same enhanced features, but it performed worse than both the Random Forest and the baseline. Since the assignment focuses on AUC, I used macro one-vs-rest AUC as the main metric for choosing the final model.

Model	Accuracy	Macro AUC
Baseline Logistic Regression	0.5210	0.9024
Random Forest with PCA + KMeans	0.5406	0.9198
Linear SVM with PCA + KMeans	0.4990	0.8896

The Random Forest model improved the macro AUC from 0.9024 to 0.9198, so I used it as my final model. The accuracy was only 0.5406, which shows that this was still a difficult multi-class problem. However, the AUC was much stronger, meaning the model was usually good at ranking the correct genre higher than the incorrect genres even when the final class prediction was not always right.

For an additional analysis, I also looked at the confusion matrix and feature importance results to better understand where the model was doing well and where it was struggling. The results also made sense when I looked at the genres individually. Classical had the strongest performance, with an F1-score of 0.86, and Anime also performed well with an F1-score of 0.77. Rap and Hip-Hop were the hardest to separate. The confusion analysis showed that 59.0 percent of true Rap songs were predicted as Hip-Hop, and 53.4 percent of true Hip-Hop songs were predicted as Rap. This makes sense because these genres can have similar values for speechiness, loudness, energy, and other audio features.

The feature importance results were also pretty interesting. Popularity was the most important feature, followed by PC1, instrumentality, danceability, loudness, speechiness, acousticness, and PC2. This showed me that the model was not just depending on one feature to make predictions. It seemed to work best when it combined popularity, individual audio features, and the larger patterns captured by PCA.

Visualization discussion

The PCA genre plot shows why this classification problem is hard. Classical separates more clearly from the other genres, which matches its strong classification score. Most other genres overlap much more in the reduced space, especially Rap, Hip-Hop, Rock, Alternative, and Electronic. The KMeans cluster plot shows that the data does have structure, but the clusters do not perfectly match the 10 genre labels. For example, one cluster was mostly Classical, while another cluster contained a large share of Rap and Hip-Hop songs. This supports the idea that the audio features capture broad musical similarity, but not always the exact genre label.

The ROC curve also supports the final model choice. All of the one-vs-rest ROC curves were above the random guessing line. Classical had the highest AUC at about 0.982, while Alternative and Rap were lower at about 0.859 and 0.891. This matches the confusion matrix, where the model had the most trouble with genres that overlap more in audio space.

Conclusion

In the end, the Random Forest model was the best model I tested. It performed better than the baseline Logistic Regression model and the Linear SVM model, with a final macro AUC of 0.9198. The hardest part of the project was not just training a classifier, but making sure the data was cleaned correctly, avoiding leakage, and making reasonable choices for PCA and KMeans. I think the model worked mainly because the audio features gave it useful information about the songs, and the PCA components helped summarize larger patterns across those features. Even though the model still confused similar genres like Rap and Hip-Hop, it was able to separate more distinct genres like Classical much more clearly.

Final AUC: 0.9198

Figures

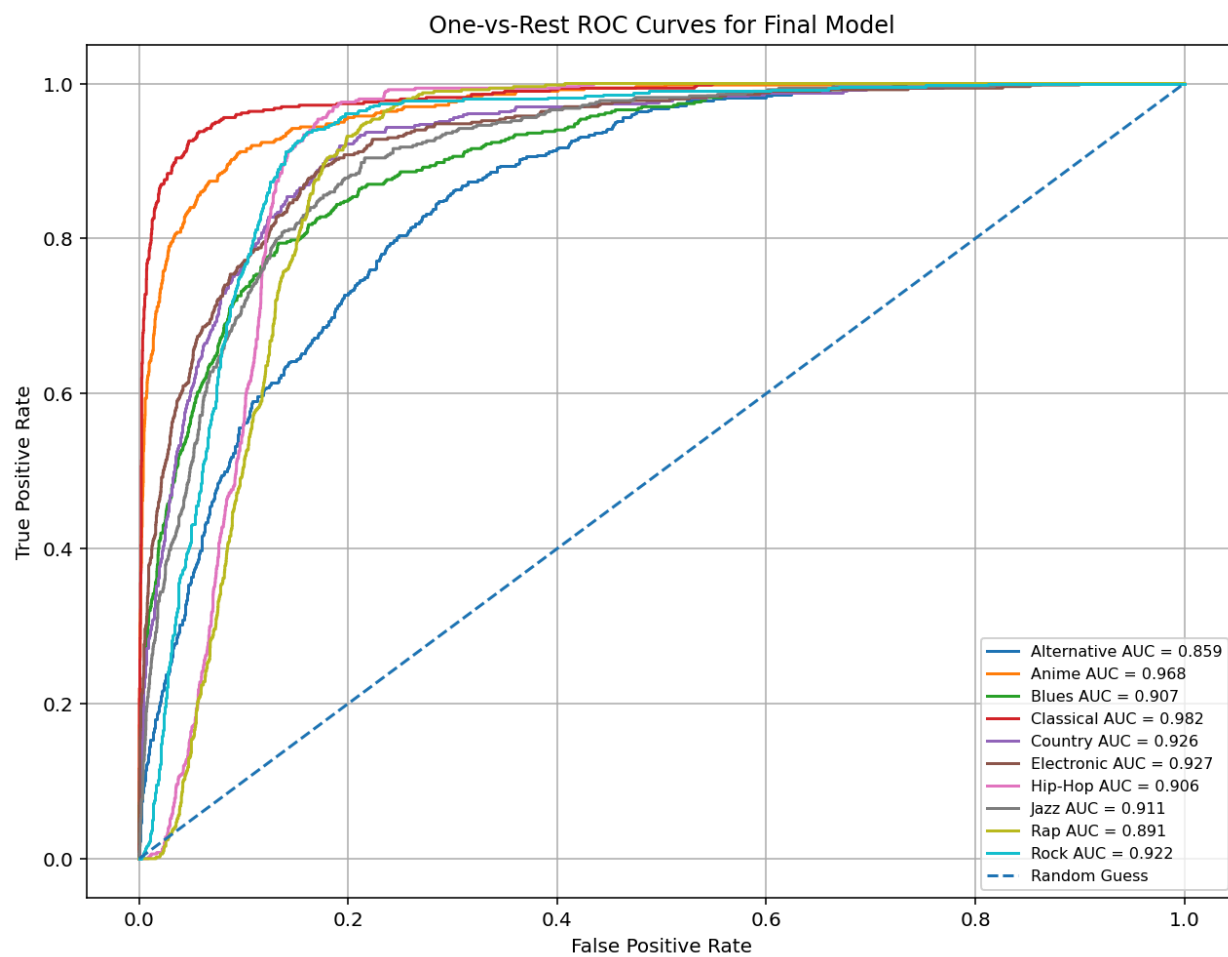


Figure 1. One-vs-rest ROC curves for the final Random Forest model.

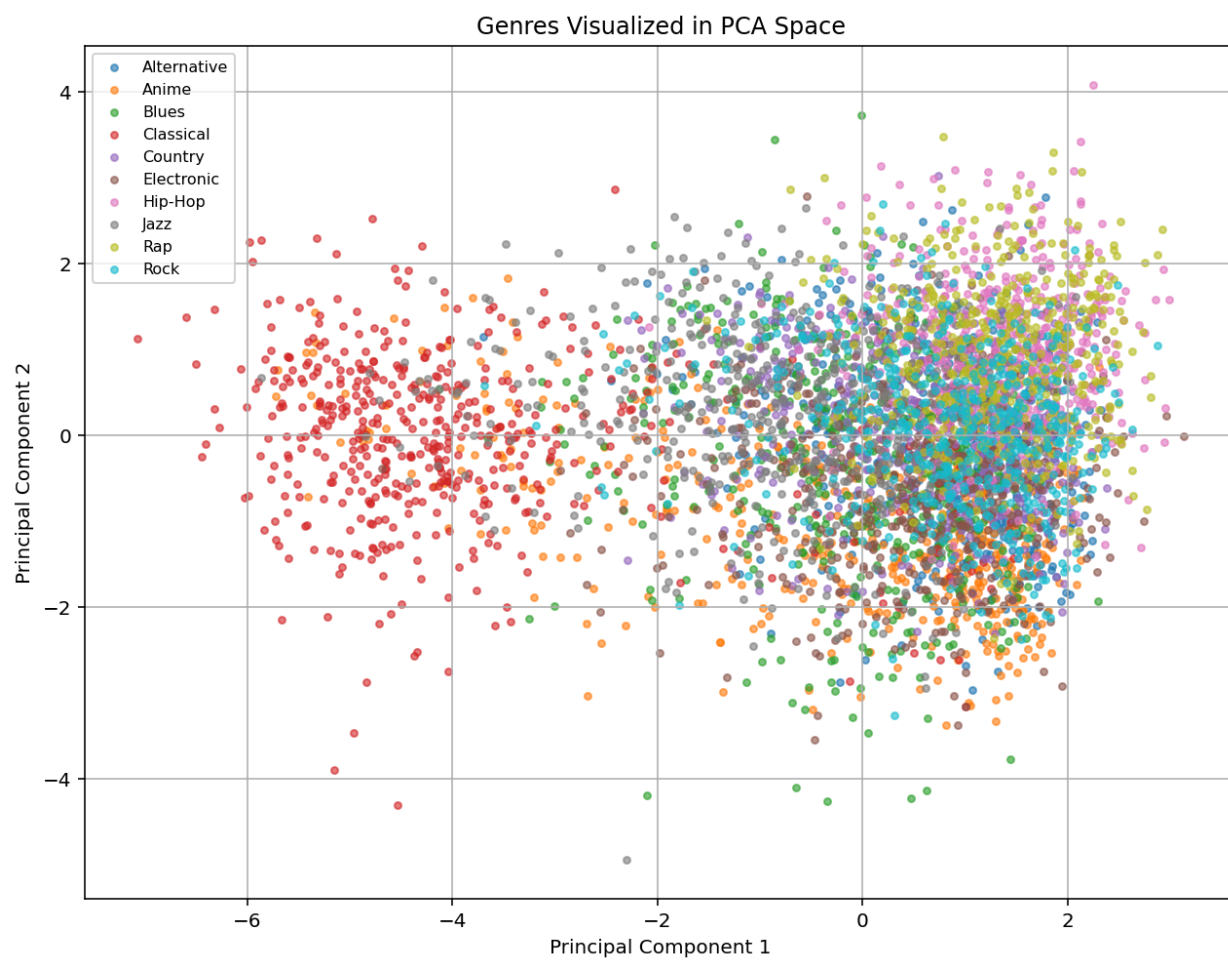


Figure 2. Genres visualized using the first two principal components.

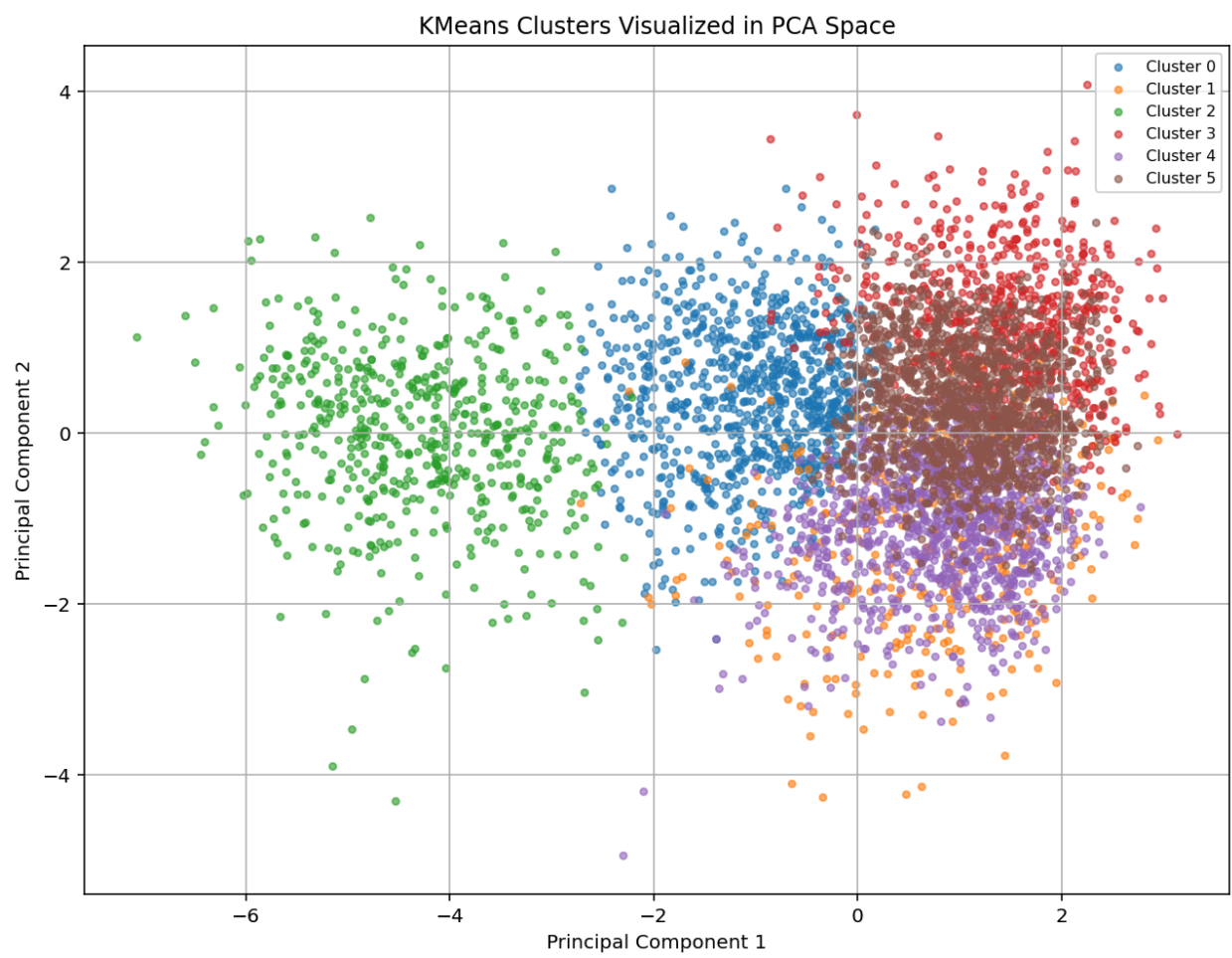


Figure 3. KMeans clusters visualized in PCA space.

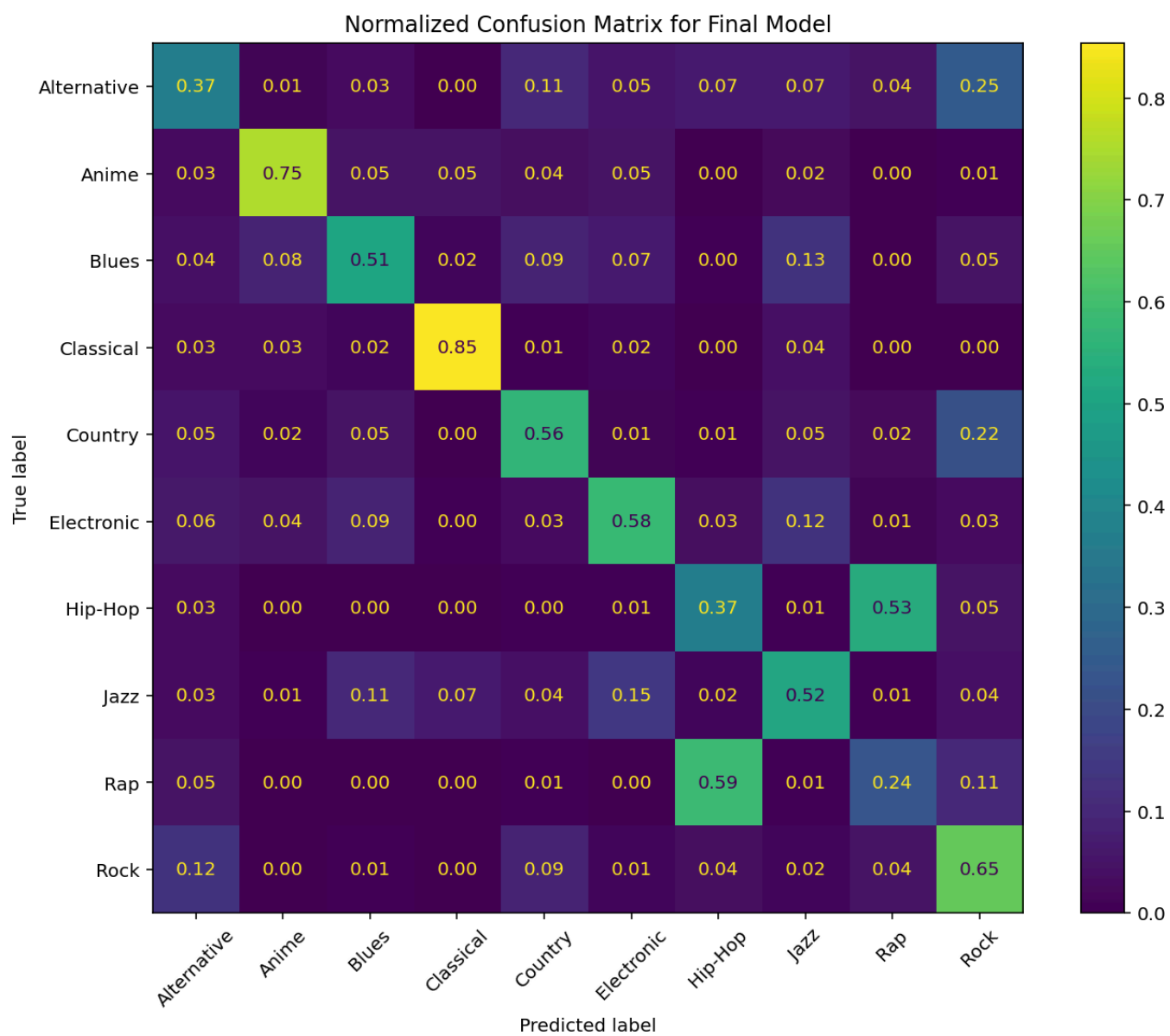


Figure 4. Confusion Matrix